DDD-W-101c
May 15, 1967
SUPERSEDING
Fed. Spec. DDD-W-101b
April 26, 1961 and
Fed. Spec. DDD-W-106
June 28, 1932

FEDERAL SPECIFICATION

WASTE, MATTED YARNS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers clean waste yarns intended for use in wiping and as packing in journal boxes.

1.2 Classification.

1.2.1 Type and grade. The waste covered by this specification shall be of one type and grade, of the following classes (see 6.2):

Class 1—Undyed, or dyed and undyed; cotton, or cotton and rayon.
Class 2—Undyed cotton only.

1.2.2 Bale or package size. Bales or packages shall be of the following sizes (weights), as specified (see 6.2):

Size 1—5-pound package.
Size 2—50-pound bale.
Size 3—100-pound bale.
Size 4—550-pound bale.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

CCC-T-191—Textile Test Methods.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The class 1 waste shall be composed of clean undyed or dyed and undyed cotton yarn or a mixture of cotton and rayon yarns. The class 2 waste shall be composed of new undyed, bleached or unbleached cotton yarns. Both classes shall conform to the following physical requirements.

3.1.1 Length of yarns. Twenty percent (plus or minus five percent of yarns), by the number of yarns present, shall be not less than 20 inches in length and not more than five percent of yarns shall be less than 3 inches in length.

3.1.2 Composition.

Cotton yarns	Class 1 (percent)	Class 2 (percent)
Undyed, bleached soft spun yarns	40 min. [1]	90 min.
Slasher yarns	20 max. [1]	10 max.
Dyed, soft spun yarn	40 max. [1]	—

[1] Twenty ±2 percent of nonbright, spun rayon yarns may be used in lieu of the same quantity by weight of cotton yarns.

3.1.3 Condition. Yarns shall be free from excessive lint, threads, hard-twisted string, shredded rag, premachined stock, soiled or oily stock, crepe yarn, lustrous rayon, yarns of more than 3-ply, dark-colored yarns, fly sweepings and dirt or other foreign matter.

3.2 Processing. The waste shall be machined twice to produce a uniformly mixed product with regard to color, type and length of yarns. The various components shall be properly floor mixed by spreading each bale of stock material on a laying pile in successive layers so that a portion cut from the edge of the pile to put through the machines will have its proper proportion of every bale of stock material used in the manufacture of waste. All spools, needles, metal clips, etc. shall be removed when found at any stage of the process.

3.3 Weight and moisture content. The waste shall meet the following requirements.

3.3.1 Weight.

3.3.1.1 Tolerance. Variation of plus or minus 10.0 percent in the gross weight of the individual bales shall be permitted, provided that the weight of the wrapping, including hoops or ties does not exceed the tare specified in table I. Any excess tare weight between 6.0 and 10.0 percent inclusive shall be deducted at the price per pound that is paid for the waste and tare. Tare exceeding 10.0 percent of the gross weight shall be cause for rejection.

TABLE I. Weight and volume

Normal gross weight of bale or package	Gross weight of bale or package [1][2]	Volume of bale
Pounds	Pounds	Cubic feet maximum
5	4.5–5.5	—
50	45–55	2.5
100	90–110	5.0
550	495–605	27.5

[1] Maximum 6.0 percent tare permitted. Tare exceeding 6.0 percent but not exceeding 10.0 percent shall be accepted on an adjustment basis as follows:
Deduction = Total gross weight of shipment × percentage of excess tare weight × price.
Where: Percentage of excess tare weight = percentage total tare − 6.0 percent. Price per pound = contract price per pound of waste.

[2] Maximum 6.5 percent moisture permitted. Moisture content exceeding 6.5 percent but not exceeding 10.0 percent shall be accepted on an adjustment (see 3.3.1.2).

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3.3.1.2 Moisture. Where the reported moisture content for a lot exceeds 6.5 percent, but does not exceed 10.0 percent, payment adjustment for the entire lot shall be made as follows:

Deduction = Total gross weight of shipment multiplied by percentage of excess moisture content multiplied by price per pound.

Where: Percentage of excess moisture = percentage moisture content present in samples − 6.5 percent.

Moisture content exceeding 10.0 percent of the gross weight shall be cause for rejection.

3.4 Extractable matter. The waste shall contain not more than 2.0 percent extractable matter when tested in accordance with 4.3.4.

3.5 Absorbency rate. Water and oil shall be absorbed by the waste within 50 seconds when tested in accordance with 4.3.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as spedified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Inspection shall be in accordance with provisions set forth in MIL–STD–105, except where otherwise indicated or specified by civil Government agencies.

4.2.1 Inspection of components.

4.2.1.1 Testing of components. In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition to the quality assurance and testing provisions of the referenced documents, components and materials shall be tested as shown in table II.

TABLE II. Testing of components

Characteristic	Requirement paragraph	Test method
Length of yarns	3.1.1	[1]
Composition		
Class 1	3.1.2	[1]
Class 2	3.1.2	[1]

[1] Supplier's certificate of compliance will be acceptable for this characteristic.

4.2.2 End item examination.

4.2.2.1 Examination of the end item. The defects found during examination shall be classified in accordance with 4.2.2.1.1 through 4.2.2.1.4 The inspection level (sample size) and acceptable quality level (AQL) shall be as specified in 4.2.2.1.5. The lot size shall be expressed in packages or bales, as applicable. The sample unit for volume and gross weights shall be one bale or package. The sample unit

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for tare weight shall be the wrapping, including hoops and ties for one bale or package. The sample unit for cleanness and condition shall be as follows:

	Sample unit [1]
5-pound packages	1.0 lb.
50-pound bales	1.0 lb.
100-pound bales	2.0 lbs.
550-pound bales	10.0 lbs.

Contents of each package or bale to be examined shall be emptied on a clean surface.

[1] Amount to be taken from each package or bale in the sample size.

4.2.2.1.1 Overall examination of the waste for defects in cleaness and condition.

Examine	Defect
Cleanness	Overall uncleanness (appreciably dirty, animal infestation, mildewed). Objectionable odor. Contains foreign matter, such as metal particles, wood splinters, paper, dirt, fly sweepings, etc. Contains soiled or oily stock.
Condition	Contains yarns other than specified, not free from shredded rag or premachined stock. Not free from threads or strings. Not uniformly mixed with regard to color (class 1) and length of yarns and not machined into homogeneous mass. Contains crepe yarn, lustrous rayon, yarns of more than three ply or dark colored yarns. The proportion of yarns are not as specified.

4.2.2.1.2 Examination for volume of bale. Any volume which does not comply with the corresponding volume specified in table I shall be classified as a defect. This examination shall be made prior to the examination indicated in 4.2.2.1.1 (this examination shall not be applicable to 5-pound packages.)

4.2.2.1.3 Examination for gross weight of bale or package. Any bale or package that exceeds the specified tolerances of table I and 3.3.1.1 or any gross weight less than the weight marked on the bale shall be classified as a defect. This examination shall be made prior to the examination indicated in 4.2.2.1.1.

4.2.2.1.4 Examination of tare weight. The total tare weight of the packages or bales taken to determine gross weight shall be obtained. If the total tare weight of the packages or bales exceed 10 percent of the gross weight of the sample, the lot shall be rejected. If the total tare weight of these packages or bales exceeds 6 percent, but does not exceed 10 percent of the gross weight of the sample, payment adjustment for the entire lot represented by the sample shall be made as indicated in table I.

4.2.2.1.5 Inspection levels and acceptable quality level (AQL's). The inspection levels and AQL's expressed in defects per hundred units shall be as follows:

For defects applicable to	Inspection level	AQL
4.2.2.1.1	I	One class 4.0
4.2.2.1.2	S–2	One class 4.0
4.2.2.1.3	S–2	One class 4.0

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4.2.2.2 Testing of the end item. Moisture content, absorbency, and extractable matter of the waste shall be performed on each lot in accordance with table III. The inspection level and acceptable quality level shall be as specified in table III. The lot size shall be expressed in packages or bales, as applicable. Each sample unit shall be a total of 2 pounds composite drawn from various portions of each package or bale taken for test. One-fourth of each sample unit shall be accurately weighed to the nearest $\frac{1}{8}$ ounce and then immediately placed in a moisture proof container and the cover of the container tightly closed, sealed, properly identified and clearly marked "FOR MOISTURE CONTENT DETERMINATION". The weight in ounces shall also be clearly marked. The container and an additional one-fourth of each sample unit shall then be submitted to the Government testing laboratory. The remainder of the sample shall be similarly weighed and packaged, but retained by the inspector for the umpire test in the event of dispute. If the average results of laboratory tests indicate that the moisture content exceeds 10 percent, the lot shall be provisionally rejected. If the supplier disputes the test results and indicates performance of a retest, the umpire test container shall be submitted to the Government testing laboratory. If the results of the retest indicate that the moisture content exceeds 10 percent, the lot shall be rejected. In all cases, if the reported moisture content exceeds 6.5 percent, but does not exceed 10 percent, payment adjustment for the entire lot represented shall be made as indicated in 3.3.1.2. The lot shall be rejected if there are any failures for excess extractable matter and if the number of sample units in the sample for absorbency exceeds the acceptance number.

TABLE III. Testing of end item

Characteristic	Requirement paragraph	Test method	Requirement applicable to		No. det. per sample unit	Results reported as	Inspection level	AQL
			Sample unit	Lot average				
Moisture content	3.3.1.2	4.3.2	—	X	2	To the nearest 0.1 percent	S–1	—
Absorbency	3.5	4.3.3	X	—	1	Pass or fail (Describe failure)	S–1	6.5
Extractable matter	3.4	4.3.4	—	X	2	To the nearest 0.1 percent	S–1	—

4.3 Testing for fiber identification and composition. Each package or bale of waste taken for tests shall be tested as described in 4.3.1 to verify conformance with the requirements of this specification. The inspection level shall be S–1 and the acceptable quality level shall be 6.5. Any bale of waste in the sample that does not conform to the specification shall be rejected, and if the number of nonconforming bales of waste in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered for Government inspection provided the supplier has repaired or removed all nonconforming bales of waste. The Government inspector shall again draw samples, when required, for both examination and tests for such resubmitted lots to verify compliance with the specification requirements.

4.3.1 Methods of testing. The tests shall be conducted in accordance with the following methods of CCC–T–191:

Identification of cotton—Method No. 1200
Identification of rayon:
Viscose—Method No. 1500
Cuprammonium—Method No. 1520

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4.3.2 Moisture content. The samples for test shall be accurately weighed and then dried in an oven at 221° to 230°F. (105° to 110°C.) until the weights from two consecutive weighings, taken not less than 5 minutes apart, differ by not more than 0.5 gram. The moisture content is the difference between the weight of the waste as sampled (Wo) and the "dry weight" (Wd), expressed as a percentage of the original (sample) weight:

$$\text{Moisture content percent} = \frac{\text{Wo} - \text{Wd} \times 100}{\text{Wo}}$$

The average of the moisture content of the individual samples shall be considered the moisture content of the entire lot represented by the samples.

4.3.3 Absorbency.

4.3.3.1 Water absorbency (rate). A 2-ounce specimen for test shall be wadded into a ball approximately 1 to 1½ inches in diameter and laid upon a nonabsorbent surface. Five drops of water at 23° plus or minus 2°C. shall be allowed to fall freely into the material from a capillary delivering 0.04 milliliter drops which is held ½ to 2 inches above the specimen. Each drop shall be applied in a different area and fifty seconds after each drop has been applied, the waste material shall be examined for absorption. Inability of the waste material to absorb the water shall be reported as a failure.

4.3.3.2 Oil absorbency. The same test procedures followed for water absorbency shall be performed on a new specimen using a light machine oil in place of the water. Inability of the waste material to absorb the oil shall be reported as a failure.

4.3.4 Extractable matter. A specimen of waste of not less than 5 grams shall be dried to constant weight at 105° to 110°C. (221° to 230°F.) and the weight recorded. The specimen shall be extracted with carbon tetrachloride in a soxhlet apparatus for 4 hours. The carbon tetrachloride shall then be evaporated from the extract and the residue dried to constant weight at 105° to 110°C. (221° to 230°F.). This residue is matter soluble in carbon tetrachloride and shall be reported as a percentage of the weight of the original oven-dried specimen.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level C only.

5.1.1 Level C. The cotton waste shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Shipping containers shall conform to carrier rules and regulations applicable to the mode of transportation.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order, each bale, package, or shipping container shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. The waste covered by this specification is utilized for:

(a) Class 1—Wiping machinery, trucks, autos and packing in journals.
(b) Class 2—For use on machinery and in laboratories.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Size (weights) required (see 1.2.2).
(c) Class required (see 1.2.1).

6.3 Soft spun yarn is defined as yarn having 20 turns per inch or less.

6.4 Supersession data. The requirements of DDD–W–106 have been incorporated in this specification and identified as class 2.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Waste, Matted Yarns - DDD-W-101c

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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